

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1551

West Frankfort 100 kV
October 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1551. The Interconnection Customer proposes a 196.3 MW solar photovoltaic facility interconnecting at the West Frankfort 100 kV substation in the Illinois area. The request was received on August 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,705,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W FRANKFORT / LUKA 100 kV
Contingency: FISK ST 230/100kV XFMR #5
Loading: 117.5% (Rating: 127 MVA)
DFAX: 48.9%
Pre-project loading: 41.8%
2. Facility: W FRANKFORT / NAVAL SUBMARINE BASE KINGS BAY 100 kV
Contingency: LUKA / W FRANKFORT 100 kV
Loading: 104.3% (Rating: 139 MVA)
DFAX: 100.0%
Pre-project loading: 37.4%
3. Facility: H ALLEN FRANKLIN COMBINED CYCLE / NAVAL SUBMARINE BASE KINGS BAY 100 kV Cir. 1
Contingency: NAVAL SUBMARINE BASE KINGS BAY / H ALLEN FRANKLIN COMBINED CYCLE 100 kV Line 3
Loading: 102.6% (Rating: 162 MVA)
DFAX: 10.4%
Pre-project loading: 90.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W FRANKFORT / LUKA 100 kV	Rebuild line with higher-capacity conductor	\$948,000	98% (\$927,000)
W FANKFORT / NAVAL SUBMARINE BASE KINGS BAY 100 kV	Reconductor line	\$625,000	100% (\$625,000)
H ALLEN FRANKLIN COMBINED CYCLE / NAVAL SUBMARINE BASE KINGS BAY 100 kV Cir. 1	Reconductor line	\$153,000	100% (\$153,000)
	Total allocated to project		\$1,705,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.